Kadi-APY Workshop

Automating everyday data management tasks in Kadi4Mat with Python and the command line

DATE AND TIME	FORMAT	MEETING LINK
Friday, 13 November 2026 **14:00-16:00 (CET)**	**Online, hands-on** **in Google Colab**	**[FILL IN]**

About the workshop

Kadi4Mat is a research data infrastructure for storing and organizing research data together with structured metadata. kadi-apy is its Python library and command line tool. It allows routine steps of the web interface, such as creating records, uploading files, adding metadata and searching, to be scripted. This is useful whenever the same steps are repeated for many files, for example after a measurement series.

In this workshop we work through such a task step by step. The example is a tensile test campaign with three steel specimens (synthetic data). The result files of the testing machine are uploaded to Kadi4Mat, characteristic values such as the yield strength are calculated and stored as metadata, and the data is searched, downloaded and plotted again. The same steps apply to other instruments and file types. No prior experience with kadi-apy or Python is required.

What participants will be able to do afterwards

- Find functions and commands in the kadi-apy documentation.
- Connect to Kadi4Mat from Python and from the command line with a personal access token.
- Create records, upload files, and add metadata (with units) and tags.
- Group records in collections and link related records, e.g. specimen and testing machine.
- Upload a folder of measurement files in one run, including values calculated from the files.
- Search records by tag, download their files and plot the data.

Schedule

The hands-on work consists of five parts with the same structure: a complete **example** that is demonstrated and run together, a **reference** of the functions used with links to the kadi-apy documentation, and an **exercise** on a similar task. Solutions are included in the notebook.

TIME	BLOCK	WHAT HAPPENS
14:00-14:10	**Welcome and demonstration**	Short introduction to the workshop and to the Kadi4Mat terms used: record, metadata, collection, tag and identifier. Live demonstration of the final result: a script creates records for three specimens from a folder of result files, and the records are then shown in the Kadi4Mat web interface. **Result:** an overview of what we build during the workshop.
14:10-14:20	**Setup check and Part 1: kadi-apy documentation**	Participants run the setup cells of the notebook (connection to Kadi4Mat, personal prefix) to check that everything works. Introduction to the documentation: the Library and CLI pages, how to read a function entry (required and optional parameters, return values), and how to get the same information inside Python (`help()`) and on the command line (`--help`). **Result:** a working connection and a way to look up any function.
14:20-14:35	**Part 2: Command line interface**	*Example:* create a record for specimen S01, upload its measurement file, add metadata with units (material, specimen length, test temperature) and a tag, and show the record. *Exercise:* create a record for the tensile testing machine with its own metadata. **Result:** two records in Kadi4Mat, created without the web interface.

14:35-14:55	**Part 3: Python**	*Example:* the same steps in Python for specimen S02: connect, get or create a record, add a list of metadata entries, upload a file, add a tag and read a value back. *Exercise:* repeat the steps for specimen S03. **Result:** a first working Python script that fills a record.
14:55-15:00	Break	Time to catch up with the exercises.
15:00-15:20	**Part 4: Batch upload of a test campaign**	*Example:* add a record to a collection and link two records (specimen and testing machine), including the difference between numeric IDs and identifiers. A loop that reads and evaluates every result file of a folder. *Exercise:* complete the loop so that for every file a record is created, the calculated values are stored as metadata, the file is uploaded, and the record is tagged, added to the collection and linked to the machine. **Result:** the complete test campaign in Kadi4Mat, uploaded with one script.
15:20-15:30	**Part 5: Search, download and plot**	*Example:* search the records by tag and print a table of their metadata. *Exercise:* download the files of the found records and plot all stress-strain curves in one figure. **Result:** the data is available again for analysis.
15:30-16:00	**Discussion**	**1. Own use cases (about 5 min):** everyone notes one repetitive data task from their own work, with instrument, file type and number of files. **2. Collecting and grouping (about 10 min):** the tasks are collected in the chat and grouped. For the most common ones we sketch on screen which kadi-apy functions would be used. **3. Open questions (about 10 min):** for example data protection when using Colab, running kadi-apy on a lab computer, importing existing data, and metadata templates. **4. Wrap-up (about 5 min):** next steps, links to the material and follow-up. **Result:** a list of concrete tasks that participants can automate next.

Before the workshop

Please complete these steps in advance (about 20 minutes). A step-by-step checklist is in the participant guide.

1. Log in to Kadi4Mat at kadi4mat.iam.kit.edu and create a personal access token (Settings > Access tokens).
2. Open the workshop notebook in Google Colab (any Google account) and save a copy to your Drive.
3. Store the token as a Colab secret named `KADI_PAT` and run the setup cells (section 0).

If something does not work, please reply to the invitation e-mail with a screenshot of the error, or join the meeting 10 minutes early.

Material

Workshop notebook	`colab.research.google.com/github/msishuvo/thematic-workshop-kadiAPY/blob/main/notebooks/kadi_apy_workshop.ipynb`
Repository	`github.com/msishuvo/thematic-workshop-kadiAPY` (participant guide, reference of the functions and commands used, example scripts, example data)
kadi-apy documentation	`kadi-apy.readthedocs.io/en/stable`